Griffiths 2021 Breast — analysis summary

RUN 1 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Samples/patients intermix per-neighbourhood on ONE connected manifold. Harmony(sample) same-sample neighbour fraction 0.91 → 0.17 (θ=5); iLISI 1.0 → 13.4 of 67 samples. ~1,100 detached cells (1.0%) removed as islands; largest component retained. Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

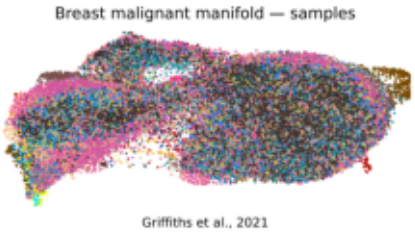
Single Gast/pSom → Crest axis, no hub diverging into two arms. Organized Gast↔pSom cross-repression (smoothed r=−0.42); Crest an inverse terminal gradient (Crest vs pSom r=−0.76; NC-high decile at pseudotime 0.77; joint pSom&Crest enrichment 0.31 < 1). Curve fits clean (2 tips, 0 forks, fold-free); PPT tree only over-fragments (5–13 forks). Crest does NOT enrich a Gast arm (no arms) — no flag.

Gate 3 — hub cross-repression → NA

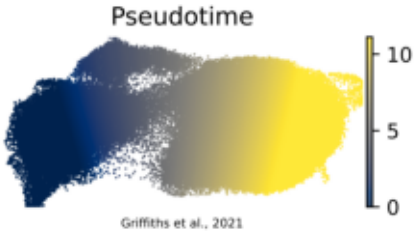
No bifurcation (non-branching geometry) → branching-only Gate 3 not evaluated; 04_correlation / 05_modules not produced (spec §6).

STEP FIGURES

Manifold — samples



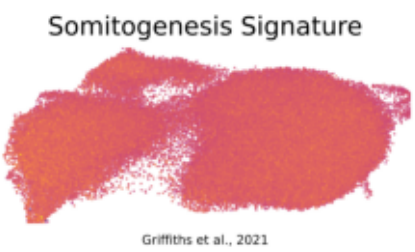
Pseudotime



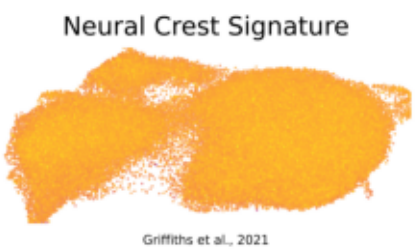
Gastrulation



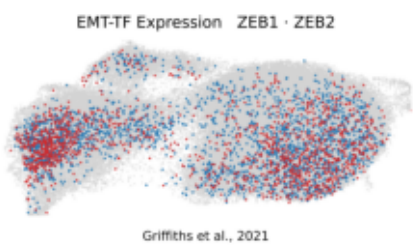
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Griffiths et al., 2021), 10x + iCell8 — 108,849 malignant cells on the trajectory (67/95 samples, 110,195 cells loaded; two technology leaves 10X 110,568 + iCell8 804, concatenated on 15,691 shared genes; a 246-cell iCell8/cycling lobe scoped out).

Verdict.
The embryonic-program pattern reproduces as the non-branching case: one continuous, sample-integrated gastrulation/somitogenesis → neural-crest axis with organized Gast↔pSom cross-repression and terminal Crest — no Gast/pSom hub bifurcating into two competing arms. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1, ZEB2 expressed but flat along pseudotime (|ρ|<0.05); overlay the two most-expressed non-directionally.

Parameter choices

Min-cell threshold	≥100 (depth-invariant floor; no dominant cliff) → 67/95 samples, 110,195 cells	**Technology**	10X 110,568 + iCell8 804 malignant, concatenated on 15,691 shared genes
Embedding	n_pcs=30, Harmony(sample) θ=5, spread=1.3, k=30	**Trajectory**	scFates curve N=20, non-branching; fold self-approach 2.95 (fold-free)
Compartment	246-cell lobe (100% iCell8, 70% cycling) scoped out as technology/cycling confound	**Root**	curve tip 0 (Gast/pSom-high, Crest-low); axis cell-cycle-free (pt vs cycling r=−0.04)
Geometry	Linear-pSom/Crest (global means Gast −0.021, pSom −0.052, Crest +0.021)	**EMT-TFs**	ZEB1, ZEB2 expressed but flat (|ρ|<0.05); overlay non-directionally; VIM excluded